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THE IMPORTANCE OF VERTICAL MOTION IN THE
DEVELOPMENT OF TROPICAL REVOLVING STORMS

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SUMMARY

The problem discussed is that of the mechanism for the eviction of air from tropical revolving storms. By the examination of the equations of motion it is found that this eviction could be explained by the upflow of air from layers of greater velocity to those of less velocity.

A proposition which may have important applications to many other problems is enumerated.

1. The whole problem of the theory of pressure changes in the atmosphere is to account for the accumulation or reduction in the mass of air above any point on the earth's surface. It is a fundamental proposition that if the winds are everywhere geostrophic there can be no change in surface pressure, and some mechanism is required to produce a departure from the geostrophic value which will cause air to be drawn away from a region of falling pressure and accumulated over a region of rising pressure. In a previous paper (Durst, 1933) one of the present authors has shown that the term in the equations of motion due to subsidence may be effective in building up and maintaining the excess pressure in a warm anticyclone. In the present paper grounds are given for believing that the same "vertical term" may be important in causing divergence above the central region of a tropical revolving storm. The clear calm bull's eye of the storm appears as an incidental outcome of the discussion.

A theory of cyclone development is described by Shaw (1922). The low pressure is given by the difference in weight between ascending saturated air and a cooler environment; by taking the whole depth of the troposphere up to 15 km. the difference in density is shown to give a reasonable approach to the low pressure actually observed. Although the pressure difference is in this way explained purely as an effect of temperatures it does not avoid the necessity of providing a mechanism for evicting the excess air. In the first place the effect on pressure changes produced by the lifting bodily of the air above is important. If the atmosphere above is lifted the isobaric surfaces are lifted with it; at any given level the pressure therefore rises, or, in other words, an isallobaric high is created, which, in accordance with the formula given by Brunt and Douglas (1928), must set up a horizontal out flow. This effect is, however, clearly insufficient, for it requires continuously rising pressure in the upper atmosphere to account for falling pressure at the surface. Shaw was therefore led to suggest a further scouring process, by which phrase one understands the dragging away of the environment by the upward moving current.

The mechanism is by no means easy to understand nor is it amenable to quantitative treatment, and a discussion in terms of the ordinary equations of motion would be more satisfactory.

2. The equations of motion of a fluid moving symmetrically round a centre may be written in cylindrical co-ordinates (Brunt, 1921) as:—

$$\frac{du}{dt} - \frac{v^2}{r} \equiv \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial r} - \frac{v^2}{r} + w \frac{\partial u}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial r} + lv + k \frac{\partial^2 u}{\partial z^2}. \quad (1)$$

$$\frac{1}{r} \frac{d}{dt}(rv) \equiv \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial r} + \frac{uv}{r} + w \frac{\partial v}{\partial z} = -lu + k \frac{\partial^2 v}{\partial z^2} \quad (2)$$

$$\frac{dw}{dt} \equiv \frac{\partial w}{\partial t} + u \frac{\partial w}{\partial r} + w \frac{\partial w}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial z} - g \quad (3)$$

where u , v , w are velocities reckoned positive in the directions r , θ , z increasing, p is pressure, k the coefficient of turbulence and l is written for $2\omega \sin \phi$, ω being the angular rotation of the earth and ϕ the latitude.

Equation (2) may be written:—

$$u \text{ (the outflow)} = \left(-\frac{\partial v}{\partial t} + k \frac{\partial^2 v}{\partial z^2} - w \frac{\partial v}{\partial z} \right) / \left(\frac{v}{r} + \frac{\partial v}{\partial r} + l \right) \quad (4)$$

The equations apply only to the motion in a stationary symmetrical circulation and are to that extent not strictly applicable to the moving cyclones which occur in nature, but there are good grounds for believing that the asymmetry and migration are complications not essential to the development. The translatory velocity of the system is generally small and the approximation to circular motion more perfect in a newly developed centre and it is considered that the treatment of the ideal case will include the essential features.

It is therefore required to estimate the magnitudes of the various factors in the above expression for the radial velocity.

3. Considering first the denominator, this may be written

$$\frac{1}{r} \frac{\partial}{\partial r}(vr) + l = \zeta + l.$$

This expression gives then the vorticity referred to axes fixed in space and is twice the local angular velocity measured positively in the cyclonic direction.

The convergence C is defined by

$$C = -\frac{1}{r} \frac{\partial}{\partial r}(ur)$$

and measures the rate at which the horizontal area covered by an element of fluid is contracting. By manipulation of equation (2) it may be shown that (ignoring friction)

$$\frac{d\zeta}{dt} = C(l + \zeta) \quad (5)$$

which is merely a special case of a general formula given by Petterssen (1936).

If, therefore, it is assumed that convergence is confined to a central area, then, outside this area, $d\zeta/dt=0$ and if the disturbance develops from a state of rest

$$\zeta=0, \text{ i.e., } \frac{1}{r} \frac{\partial}{\partial r}(vr)=0$$

i.e., vr is a function of t only.

This is the well-known result that the withdrawal of air from a central region produces a vr vortex in the surrounding region.

The distribution of vorticity within the central region depends on how the convergence is distributed, but it is clear from equation (5) that ζ increases where convergence takes place. It therefore follows that $\frac{1}{r} \frac{\partial}{\partial r}(vr)+l$ is greater than l in the region of convergence but approximates to l where convergence is negligible. This theoretical conclusion is borne out by the observations made at Hong Kong during the passage of a typhoon on August 18th, 1923. Commander Hennessy* gave the values of the wind speed at various distances from the centre and called attention to the close relationship between the velocity and the inverse of the radius. The observations are set out in Table I. The approximate constancy of the product vr is shown clearly in column 3. The closeness to the vr vortex depends on the relative magnitudes of $\frac{1}{r} \frac{\partial}{\partial r}(vr)$ and the constant l which in the latitude 20°N. is 0.18 radians per hour. Estimates of $\frac{1}{r} \frac{\partial}{\partial r}(vr)$ are obtained from the differences of successive values of vr and are shown in column 4.

TABLE I

1	2	3	4
Distance from centre r km.	Surface wind velocity v m./s.	vr $m^2/s \times 10^5$	Vorticity $\frac{1}{r} \frac{\partial}{\partial r}(vr)$ radians/hour
149	5.4	8.0	} -0.22 +0.10 -0.13 +0.17 +0.02 -0.18 +0.25 -
128	7.6	9.7	
106	8.5	9.0	
87	11.2	9.7	
64	13.9	8.9	
45	19.7	8.8	
27	33.5	9.1	
22.5	37.1	8.3	
0	0	0	

Between the centre, where the velocity is of necessity zero, and $22\frac{1}{2}$ kms. $\frac{1}{r} \frac{\partial}{\partial r}(vr)$ must be large in comparison with the values at greater distances.

* In the *Marine Observer*, August, 1924. The wind speeds are there given in mi./hr. and the distance from the centre in miles. These are converted to m./sec. and kilometres, but the computations for columns 3 and 4 are made directly from the observations as given.

It is observed that these magnitudes at more than 27 km. from the centre show erratic results on either side of zero such as may be expected from the nature of the observations, but the average value is small compared with 0.18; the assumption of a vr vortex is therefore justified. Within the radius of 27 km., however, the value becomes definitely positive and large compared with l as is to be expected as a result of convergence; the result suggests that most of the horizontal convergence due to convection takes place within this inner core.

Any attempt to estimate the denominator in equation (4) is therefore not likely to be very successful; it is at least l , but may be considerably greater. We are, however, in the present discussion, concerned mainly with the relative importance of the three terms in the numerator.

4. The origin of the storm by vertical instability, owing to a suitable distribution of temperature and humidity, is assumed. The low pressure is then initially due to the higher temperature of the ascending air, the lifting bodily of the air above causes isallobaric outflow, and inflow is started at the surface. The term $\partial v/\partial t$ which is due to the general development, and may be called the isallobaric term, gives inflow at the surface but can only give outflow above if $\partial v/\partial t$ is negative, that is, if anticyclonic circulation is developing and pressure rising at the centre in these upper layers. Such a process presumably exists in the early stages, but clearly cannot persist, for pressure does not continue to rise in the upper atmosphere while it falls at the surface.

The term $k \partial^2 v/\partial z^2$ is the frictional term and is mainly effective in increasing the inflow near the surface. Its value is presumably negligible in the upper atmosphere and cannot be instrumental in providing the necessary outflow above.

If, therefore, it is possible to account for the development in the nature of a symmetrical convective system the only possible cause for the outflow above must be in the term $w \partial v/\partial z$. Since the circulatory velocities decrease with height as the pressure gradients decrease $\partial v/\partial z$ is negative above the surface turbulent layer while w is positive and increases with height if convergence is taking place below; it follows that this term will give a possible explanation of the outflow. The physical explanation is that the air having a velocity appropriate to its level is carried upwards into a region of less pressure gradient, its velocity is then in excess of the value required for circular motion and it is thrown out from the centre. It remains only to show that the order of magnitude of the term is sufficient to be significant.

The magnitudes of the three terms will now be considered in turn:—

$$(a) \quad -\frac{\partial v}{\partial t}; \text{ the isallobaric term.}$$

The maximum velocities near the centre of the storm are of the order 50 m./sec. and if the storm takes 24 hours to develop, $\partial v/\partial t$ is of the order

$$\frac{50 \times 100}{24 \times 60 \times 60}$$

or 0.06 cm./sec.².

This may therefore be taken as a maximum value attained near the surface and near the centre; it will decrease with height and with distance from the centre.

$$(b) \quad k \frac{\partial^2 v}{\partial z^2}; \text{ the frictional term.}$$

Both k and $\partial^2 v / \partial z^2$ are difficult to estimate and only a broad order of magnitude can be expected. If the wind increases by 20 m./sec. in 1,000 m. through the turbulent layer and decreases above that height then $\partial^2 v / \partial z^2$ is of the order

$$\frac{20}{1,000 \times 500 \times 100} = -4 \times 10^{-7}.$$

Taking k as 10^5 the term becomes -0.04 cm./sec.² in the turbulent layer; it may be assumed to decrease rapidly with height and with distance from the centre.

$$(c) \quad -w \frac{\partial v}{\partial z}; \text{ the vertical term.}$$

It will be sufficient, for the purpose of deciding whether this term is quantitatively adequate to account for the outflow, to show that its magnitude may be of the order 0.1 cm./sec.².

The magnitude of $\partial v / \partial z$ depends on the vertical extent of the storm and if it is assumed to extend through the whole troposphere, say 15 km., then with a surface velocity of 50 m./sec. $\partial v / \partial z$ is of the order

$$\frac{50}{15,000} = 3 \times 10^{-3} \text{ sec.}^{-1}.$$

We therefore require w to be of the order 0.3 m./sec.

Consider the equation of continuity applied to the central region of 50 km. diameter and 1 km. height; a vertical velocity of 0.3 m./sec. averaged over the upper surface requires an average rate of inflow of about 4 m./sec. The rate of inflow near the surface is often much in excess of this as is known from observation while convergence presumably extends up to if not beyond 1 km. A vertical velocity of 0.3 m./sec. is therefore by no means a large estimate.

An estimate of w may also be obtained from rainfall measurements; for if it is assumed that rising air is saturated and that the moisture falling as rain is replaced by the moisture content of the air entering below, then at a temperature of 300°A . (absolute humidity 24 gm./m.³) an ascending velocity of 0.3 m./sec. corresponds with a rainfall of 26 mm./hr.

The intensity of rainfall is often in excess of this while the assumption of saturation at 300°A . (81°F .) is probably a high estimate of the moisture content.

There are therefore ample grounds for believing that the vertical term is sufficiently large to account for the required eviction of air.

The actual value of the inflow and outflow computed from equation (4) depends as we have seen on the uncertain value of the denominator which has a minimum value l but is considerably greater near the centre.

Taking the numerator as 0.1 cm./sec. negative due to $\partial v/\partial t$ and $k \partial^2 v/\partial z^2$ near the surface and positive due to $w \partial v/\partial z$ in the upper air where the other terms become small, then with l as the denominator the range in the inflow or outflow is

$$\pm \frac{0.1 \times 60 \times 60}{0.18 \times 100} = \pm 20 \text{ m./sec.}$$

This is presumably excessive, but the larger value of the denominator in the region of convergence would reduce this to a reasonable value.

5. In order to complete the picture of the circulation near the surface, conditions in the immediate vicinity of the centre require closer attention. Here observation shows an area of light winds and often clear skies surrounded by a ring of maximum velocity. The necessity for such a structure may be demonstrated as follows:—

Reverting to equations (1) and (2) in the form:—

$$\frac{du}{dt} = \frac{v^2}{r} + lv - \frac{1}{\rho} \frac{\partial p}{\partial r}$$

$$\frac{1}{r} \frac{d}{dt}(rv) = -lu$$

that is neglecting friction, it is at once seen that

$$rv = -l \int u r dt = -l \int \frac{dr}{dt} r dt = \frac{l}{2} (r_0^2 - r^2) \quad (6)$$

where r_0 is the distance from which the air has been drawn.

$$\frac{du}{dt} = \frac{l^2 (r_0^2 - r^2)^2}{4 r^3} + \frac{l^2 (r_0^2 - r^2)}{2 r} - \frac{1}{\rho} \frac{\partial p}{\partial r}$$

The first two terms of the right hand side increase without limit as the centre is approached and must ultimately overcome the effect of the pressure gradient unless this also increases without limit. There is, however, no mechanism for producing a continuous increase in gradient right to the centre and indeed near the centre it is known from observation that the gradient decreases. It is clear then that there must be an inner core beyond which inflow cannot extend and on the edge of which the inflowing air has attained its highest velocity. Within this core there can be no further convergence. There must in fact be some slight divergence for the rapidly moving air just outside the core will constantly give a velocity to the outer edge which is unbalanced in respect of the pressure gradient and consequently some air is drawn out of the core.

6. On the basis of the above discussion a schematic representation of the structure of a well developed tropical revolving storm has been attempted in Fig. 1 which shows the lines of flow in a vertical section through the centre. The inflow near the surface is the result of the isallobaric and frictional terms and is limited at the central core by the dynamical considerations of paragraph 5. The convergence below causes a gradually increasing vertical velocity, the term $w \partial v/\partial z$ then becomes more important than the other terms which decrease with height; there is a consequent outward acceleration and the air rises along an inclined

path diverging above, providing the returning portion of the circulation and preventing the filling in of the low pressure from below. Since there can be no surface convergence at the core itself there is no upflow there and the falling pressure cannot be accounted for as the effect of low density of rising saturated air. It is therefore suggested that the low pressure is communicated to the core by the divergence of the ascending air which draws air away from the centre, producing some subsidence and adiabatic warming.

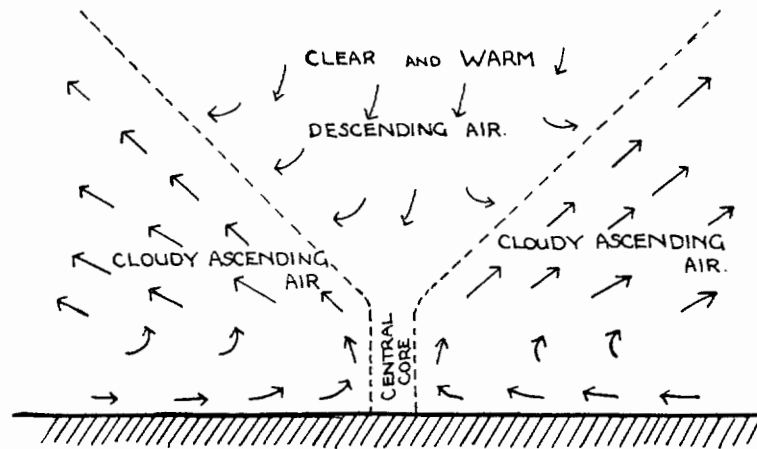


FIG. 1—Sketch of section through tropical revolving storm (stationary).

7. The authors believe that the discussion of the effects of vertical motion in the particular cases of the warm anticyclone and the tropical storm are very suggestive and may lead to important inferences in other problems of pressure changes.

It has frequently been stated that the terms due to vertical motion in the general equations may safely be ignored, but if these results are correct the position needs to be reviewed.

In the theory of the variation of geostrophic winds with height it is shown that to a close approximation the change is directed parallel to the isotherms and in the cyclonic sense with reference to the lower temperature—the so-called thermal wind. The equations of motion referred to rectangular axes are:—

$$lv' = \frac{du}{dt} \equiv \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z}$$

$$-lu' = \frac{dv}{dt} \equiv \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z}$$

where u' and v' are the components of the departure of the wind from its geostrophic value. There is therefore a component due to vertical motion directed at right angles to the vector change of wind with height or approximately at right angles to the isotherms if the winds are nearly geostrophic. The significance of the vertical term may therefore be stated in the form of a general proposition:—

"If vertical motion occurs in the atmosphere in a region of horizontal temperature gradient then ascending motion gives a component of wind from warm to cold and descending motion a component from cold to warm."

This component which gives a departure of the wind from its geostrophic value is quite distinct from that due to other causes such as, for example, the isallobaric term.

The authors have been unable to find any indication of this proposition in the literature; it is obvious that qualitatively it would throw light on many points which are at present obscure, such as: the intensification of anticyclones; the effectiveness of a cold front in causing the collapse of a warm anticyclone; the deepening of occluding depressions and the low stratosphere associated with them, as well as the structure of fronts. The success in the treatment of the special cases justified a thorough examination of the general application, which the authors hope to present at a future date.

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DISCUSSION

Sir Gilbert Walker, while expressing appreciation of the importance of the subject, regretted that there were portions of the paper with which he did not agree.

In section 2 of the paper we had three equations for the variations of u , v , w respectively; but we had no equation of continuity, and stranger still none for the variations of density on which the whole problem turned; without equations involving both these conditions, five equations in all, an effective mathematical treatment seemed to him impossible. But the authors did not consider all the three equations to which they limited themselves; they picked out one equation and ignored the rest as not essential. For example, a change in the sign of g and supposing that gravity acted upwards instead of downwards, would not affect the analysis. Further we had no mathematical treatment of the boundary conditions between the cyclone and the undisturbed air above it.

Since the equations were valid a single one selected at random would presumably give a correct value of u if actual values of the other variables were substituted; but this process could scarcely be looked upon as giving an explanation of the phenomena.

Regarding the discussion in section 5 of the conditions near the centre, it seemed rather paradoxical, after having obtained an equation for v which demonstrated that that velocity became infinitely large at the centre, to deduce that there was there "a region of light winds." Did the authors regard that equation as correct?

Prof. D. BRUNT said he was interested mainly in the physical assumptions underlying the mathematical treatment in the paper. The theory put forward by the authors was essentially a convective theory, which led to the result that, provided sufficient convection took place, a cyclone was formed. He had always imagined that in tropical regions the necessary degree of instability was to be found over a wide area, and that the reason why tropical cyclones were not more frequent was to be found in the necessity for some second condition to be fulfilled, in addition to the instability. He would therefore like to ask the authors why, if his interpretation of their theory was correct, tropical cyclones were not more frequent. (*Added later.*—Convection on a large scale appears to be more frequent over land than over the oceans in the tropics, and generally leads to the formation of thunderstorms.)

With regard to the table of velocities given in the table, he asked whether the distances were measured from the centre of isobars, or from some centre of winds, or centre of revolving fluid. An error in fixing the centre would affect the value of the product vr to a greater extent near the centre than at greater distances, where the velocity was small.

Prof. Brunt expressed his disagreement with the theorem in section 7. The theory as stated refers to an unchanging pressure distribution. Any deviation from the geostrophic velocity will lead to an uncompensated deviating force at right angles to the vectorial difference between the actual and the geostrophic wind. The deviations from geostrophic wind at any level are here assumed to be the result of motion in the vertical. Assume for simplicity that the isotherms between 5,000 and 6,000 feet run north to south, with low temperatures to west. Then the wind at 6,000 feet differs from the wind at 5,000 feet by an added component, the thermal wind, blowing in the south to north direction. If a mass of air be displaced from 5,000 to 6,000 feet, it will arrive there without the added thermal component, and will have a component of velocity relative to its environment directed towards south. In virtue of the fact that there is no pressure gradient to compensate for this, the air will have a deviating acceleration to the right, i.e., towards west, but its velocity will be towards south, along the mean isotherms. This is consistent with the mathematics of section 7, but not with the interpretation there given.

Prof. Brunt also referred to the expression "manipulation of equation (2)," and suggested that the working out of the following equation should be shown in full.

Flt.-Lieut. R. G. VERYARD pointed out that one of the main features of a tropical revolving storm was an enormous sheet of cirro-stratus cloud. He did not see how that could be accounted for on the basis of the authors' diagrams.

The AUTHORS (in reply, partly communicated later) admitted that Sir Gilbert Walker's remarks would be fair criticism if the object had been to produce a complete mathematical solution of the problem, but no attempt has been made to obtain a complete solution; their main concern was with a particular aspect of the motion—the mechanism whereby the air converging near the surface might be evicted at higher levels. There was some advantage in publishing the three equations of motion for purposes of reference; that only one should be required for their argument

was a fortunate circumstance upon which the authors felt they might congratulate themselves; it could hardly be held to invalidate the conclusion.

While the equations of continuity and the distribution of density were not specifically stated their importance to the problem was an essential part of their argument; indeed the equation of continuity leads to the problem the authors had set themselves, while the variation of density is essential to the decrease of wind with height which is postulated.

Equation (6) certainly shows that if in-flowing air did penetrate to the centre the circulatory velocity would be infinitely large, but their further argument was intended to show that owing to this tendency there was no process whereby air could be made to penetrate beyond a certain finite distance and that this limitation could account for the central core of little disturbance.

In reply to Professor Brunt the authors agreed that their contribution was based on the assumption of a convective origin for the storm, although their result should apply to any symmetrical vortex, whatever its origin, provided the circulatory velocity falls off with height. Why tropical storms should develop so rarely was an awkward fact in any theory and no doubt some form of trigger action was essential.

As regards the proposition of section 7, the argument given by Professor Brunt was at first sight convincing, but the authors feel that a more critical examination would confirm their own result. The equations of section 7 were the precise mathematical equivalent of Professor Brunt's statement that "any deviation from the geostrophic velocity will lead to an uncompensated deviating force at right angles to the vectorial difference between the actual and the geostrophic wind."

The vectorial difference due to vertical motion has by these equations the components

$$lv' = w \frac{\partial u}{\partial z}, \quad -lu' = w \frac{\partial v}{\partial z}$$

and the effect is clearly at right angles to the vector change of wind with height (approximately the thermal wind). The authors could not see how Professor Brunt reconciled his argument with the mathematics.

Vertical motion may be regarded as transferring air from one pressure field to another with an effect similar to that of imposing an isallobaric field with the isallobars parallel with the isotherms: the geostrophic departure is at right angles rather than along the isotherms. Professor Brunt's argument would be valid presumably for an air mass rising rapidly through an environment, but not for the case of general and gradual ascent with which we are concerned in large scale motion. The authors still believed that so far as it went section 7 was quite sound.

The diagram given was of course only intended to illustrate the argument and was not drawn to scale, but there seemed no reason why the cirro-stratus cloud mentioned by Mr. Veryard should not appear in the upper levels of the cloudy ascending air.